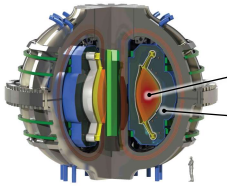
LIBRA Liquid Immersion Blanket - Robust Accountancy

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arpa·e OPEN 2021

Tritium breeding

2030



The ARC concept relies on the Liquid Immersion Blanket (LIB) for breeding tritium

$$D + T \rightarrow He + n$$

$$Li + n \rightarrow T + He$$

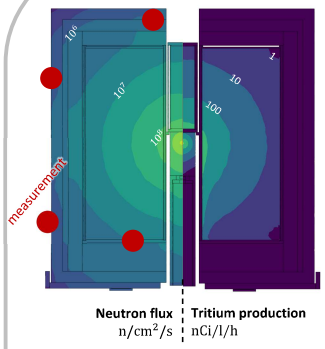
$$\text{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Can we test this concept now?

Goals of LIBRA

- ✅ To demonstrate the concept of LIB (TBR ≈ 1)
- To improve the Technology Readiness Level of LIB
- To validate numerical models for tritium transport
- To gain knowledge on molten salt behaviour in fusion environments

Neutronics



- Neutronics simulations track the path of neutron particles
- Used to predict tritium generation, neutron flux, heating, production of activated materials...

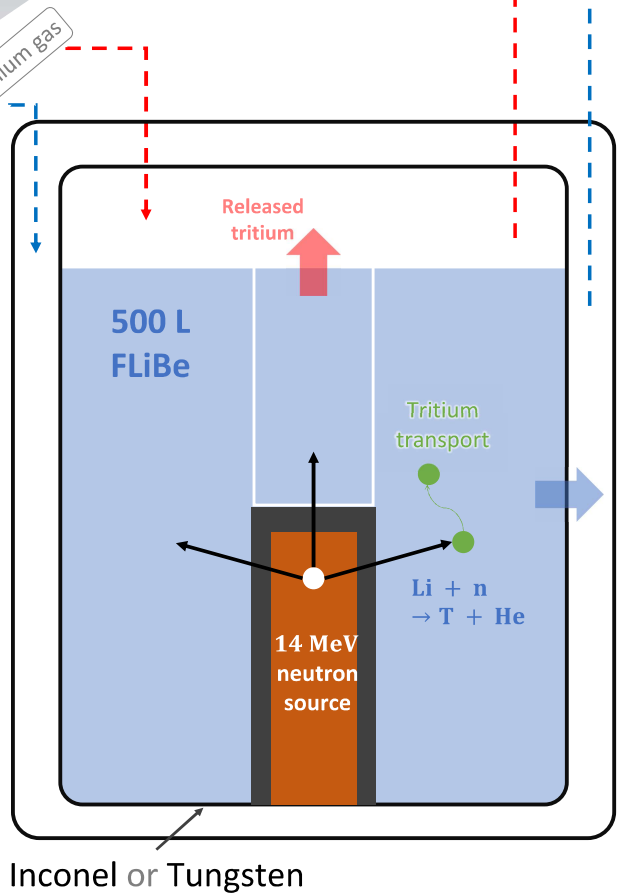


Activation foils

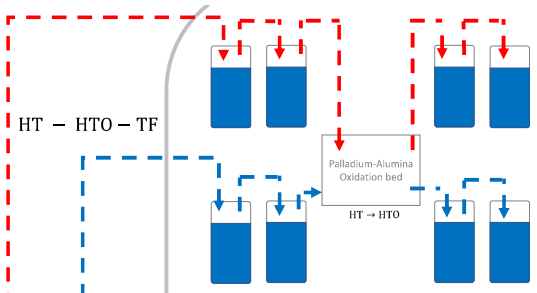
- LIBRA will provide neutronics data for validating codes
- Activation foils and diamond detectors as used to characterize the neutron source and the experiment



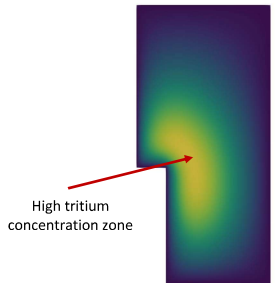
Diamond detection



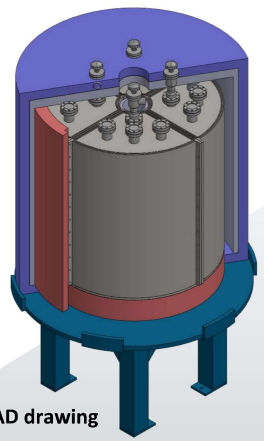
Tritium detection



- Bubblers will recover 99% of the tritium
- Water samples will be analysed with Liquid Scintillation Counting
- Ion chambers will also provide real-time measurements of tritium concentration in the gas stream
- LIBRA will provide data for simulation codes validation
- The impact of sparging bubbles in the FLiBe will also be studied



Tritium transport can be modelled at full scale



CAD drawing





LIBRA specs

Salt volume	500 L
TBR	≈ 1
Temperature	650 °C
Neutron energy	14 MeV

Higher fluid temperature = Higher Efficiency

